Cyber Risk Assessment Platform using Python

- An Intelligent System for Automated Threat Detection and Risk Analysis

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Introduction

- Cyber threats are increasing rapidly.
- Manual analysis is slow.

- This system automates:
- - Vulnerability detection
- - Threat analysis
- - Risk assessment

Problem Statement

- Organizations struggle with:
- - Identifying vulnerabilities
- - Prioritizing threats
- - Real-time monitoring

- Solution: Automated system

Objectives

- - Scan network vulnerabilities
- - Integrate threat intelligence
- - Calculate risk scores
- - Display results in dashboard

System Modules

- 1. Vulnerability Scanner
- 2. Threat Intelligence
- 3. Risk Scoring
- 4. Dashboard

Methodology

- Input IP → Scan → Analyze → Risk Score → Display Output

Technologies Used

- Python, Flask, Nmap
- HTML, CSS, JavaScript
- Optional APIs

Results

- Detects open ports
- Analyzes threats
- Shows risk levels
- Displays results clearly

Conclusion & Future Scope

- Improves efficiency
- Reduces manual work

- Future:
- - Alerts
- - API Integration
- - Cloud deployment